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A VideoMAE-v2 Approach to Zero-Shot Traffic Accident Anticipation

Team BUPT MIC Lab
Siyuan Li[1], Xiaoyang Bi[1], and Mengshi Qi[1*]
[1]State Key Laboratory of Networking and Switching Technology
[2]Beijing University of Posts and Telecommunications
{lisiyuan0818, bxy, qms}@bupt.edu.cn

Abstract

Traffic accident anticipation—predicting the likelihood of an imminent collision at every frame of a dashcam video—is safety-critical yet difficult to scale, because collecting in-domain annotated accident footage for every deployment scenario is prohibitively expensive. We study this task under a zero-shot *setting where no target-domain training data is available: the model must learn exclusively from a publicly available binary-labelled driving-accident dataset and generalise to unseen dashcam footage. We propose a framework that bridges the gap between the frame-level temporal risk estimation task and coarsely labelled binary accident datasets by coupling a VideoMAE-v2 backbone with a per-frame prediction head under a sliding-window protocol. Our method achieves* ***2nd place*** *in the 2026 CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition [1]. Code is available at* https://github.com/TimeSouth/zero-shot-taa-solution.

1. Introduction

Traffic accident anticipation (TAA) aims to estimate, at every frame of a dashcam video, the probability that a collision or near-miss will occur in the immediate future. Accurate and timely frame-level risk prediction is a prerequisite for driver-assistance alerts and autonomous emergency braking [2, 5], alongside other safety-critical perception tasks in robust autonomous driving and 3D scene understanding [6, 8, 19, 20, 23, 25]. Prior work has achieved encouraging results when in-domain annotated data is available [2, 3, 5, 22]; however, collecting and labelling dashcam accident footage for every target domain is prohibitively expensive. A more practical paradigm is *zero-shot* TAA, in which the model is trained solely on publicly available driving-accident data and must generalise to previously unseen dashcam distributions without any target-domain supervision.

Under this zero-shot setting, two principal technical challenges arise. First, a *task–data granularity mismatch*: TAA requires dense, frame-level risk scores over clips of a fixed temporal format (*e.g.*, 150 frames at 30 fps), yet publicly available accident datasets provide only coarse binary labels (accident vs. normal). Bridging this granularity gap during training is essential for meaningful per-frame supervision. Second, a *long-clip / short-window modelling* problem: while advanced video representation and multimodal learning have driven significant progress across various visual tasks (e.g., action assessment, temporal localization, video retrieval, and commonsense reasoning) [10–17, 24], state-of-the-art video encoders such as VideoMAE-v2 [21] accept only 16 frames per forward pass, substantially fewer than the full clip length, necessitating a principled window-to-clip aggregation mechanism.

We address the first challenge by restructuring a publicly available binary-labelled accident dataset into temporally standardised clips with explicit positive–negative balancing, so that per-frame supervision can be derived from clip-level annotations (Sec. 2.2). We address the second by coupling a VideoMAE-v2 backbone with a per-frame prediction head under a sliding-window protocol, and recovering dense clip-level risk sequences via overlap averaging and temporal interpolation (Sec. 2.3). Furthermore, we observe that the source-trained model suffers from a systematic *posterior shift* when applied to the target domain: the relative ranking of frame-level predictions is well-preserved, yet the predicted confidence distribution is systematically compressed, preventing effective decision-making. We address this with a lightweight, training-free test-time domain adaptation module that leverages task-specific temporal priors and the model's own prediction-distribution statistics to recalibrate the output without any target-domain labels (Sec. 2.4).

Our method achieves **2nd place** in the 2026 CVPR@AUTOPILOT Zero-Shot Accident Anticipation

*Corresponding author

competition [1]. Our design choices are further validated by two ablations (Sec. 3) demonstrating that two seemingly attractive alternatives—*late-window classification* and *cross-dataset training without class balancing*—both lead to significant performance degradation.

2. Method

2.1. Task Definition and Notation

Given a dashcam video clip of T frames, traffic accident anticipation (TAA) requires producing a per-frame risk sequence $\{p_t\}_{t=1}^{T}$, where $p_t \in [0, 1]$ denotes the estimated probability that a collision or near-miss will occur *after* frame t within the same clip. In the zero-shot setting studied here, no target-domain training data is available; the model is trained on an external source dataset (*e.g.*, Nexar [9]) in which each video carries only a clip-level binary label $y \in \{0, 1\}$ (accident vs. normal) and the event onset time τ. During training, we derive frame-level supervision $y_t \in \{0, 1\}$ from these clip-level annotations (detailed in Sec. 2.2).

Our pipeline consists of three stages: (i) *dataset curation* (Sec. 2.2), which restructures the source data into temporally standardised clips with balanced class distribution; (ii) *model training and inference* (Sec. 2.3), which finetunes a video encoder to predict $\{p_t\}$ via a sliding-window protocol; and (iii) *test-time domain adaptation* (Sec. 2.4), which recalibrates the output distribution at test time by exploiting task-specific temporal priors and the model's own prediction-distribution statistics, without requiring any target-domain labels.

2.2. Dataset Construction

Source. We use the public Nexar Crash / Dashcam Collision Prediction dataset [9] as the sole training and validation source. Each video is annotated with a clip-level label (collision / near-miss vs. normal driving) and the time-of-event τ.

Clip Splitting. Each competition [1] test clip is exactly 150 frames at 30 fps, so we slice every Nexar video to match. For positive videos we discard the last 2 s before τ, then slide a 5 s window over the remaining footage, so the model never observes the collision moment itself. For negative videos we slide directly and set $\tau = 250$ (well outside the clip) so that the time-weighted CE term reduces to plain CE. Frame-level labels follow $y_t = 1$ in positive clips and $y_t = 0$ otherwise.

Class Balancing. A naïve sliding-window split is heavily biased toward negative clips, on which a CE objective collapses to the trivial "always-negative" minimiser. We therefore apply a tiered balancing scheme that (i) keeps every positive clip, (ii) keeps every negative clip cut from a positive video—these provide informative *pre-/post-event* context, and (iii) sub-samples clips from the purely-negative videos breadth-first (at most three per video) until the positive : negative ratio is roughly 1:1. Capping per-video rather than uniformly sub-sampling preserves diversity in scenes, weather and camera setups, which we found to be important for cross-domain generalisation.

Input Preprocessing. We down-sample 30 fps to 10 fps (frame interval of 3), so a 5 s clip becomes 50 frames; we then extract overlapping 16-frame windows with stride 4 to match the VideoMAE input. Frames are resized to 224×224 and ImageNet-normalised. Training augments with horizontal flip ($p = 0.5$) and brightness/contrast jitter ($p = 0.3$).

2.3. Training and Inference

Architecture. We adopt VideoMAE-v2 Base [4, 18, 21] as the video encoder. Each 16-frame window $X \in \mathbb{R}^{16\times3\times224\times224}$ is encoded into a set of spatio-temporal tokens $Z \in \mathbb{R}^{N\times D}$. On top of the encoder we attach a lightweight per-frame classification head that fuses a global mean-pooled feature with temporally upsampled token features, producing per-frame risk probabilities $\{p_t^{(w)}\}_{t=1}^{16} \in [0, 1]$ for each window. The entire model is trained end-to-end.

Training objective. For window samples with frame label $y_t \in \{0, 1\}$, time-to-event τ and global frame index t, we minimise

$$\mathcal{L} = \tfrac{1}{2}\Big(\mathbb{E}_{y_t=1}\big[e^{-\max(0,\,(\tau-t-1)/\mathrm{fps})}\cdot\mathrm{CE}\big] + \mathbb{E}_{y_t=0}[\mathrm{CE}]\Big) \tag{1}$$

This is the discrete "Exp-Loss" commonly used for accident anticipation [2]: it down-weights frame predictions that are far from the event so that the model is encouraged to be confident only when the accident is temporally imminent, while still being penalised by full CE elsewhere.

Window-to-clip aggregation. For each 150-frame test clip we (i) slide a 16-frame window with stride 1 across the 10 fps-sampled sequence, (ii) run the network on every window to obtain $\{p_t^{(w)}\}$, (iii) for every sampled frame position average the predictions of all windows that cover it (*overlap averaging*), and (iv) linearly interpolate the resulting sparse probability vector back to all 150 frame indices (*temporal interpolation*). Step (iv) is necessary because the model only outputs predictions at 10 fps-sampled positions, while the evaluation requires 30 fps frame-level scores.

2.4. Test-Time Domain Adaptation

We observe that the source-trained model suffers from a *posterior shift* on the target domain: frame-level risk rankings are well-preserved, yet the absolute confidence scale is systematically compressed, a common artefact in unsupervised domain adaptation. We address this with a training-free, three-component test-time adaptation module.

Temporal confidence aggregation. In the TAA task, frames closer to the end of a clip carry stronger causal evidence about an impending event, since the accident—if any—occurs shortly after the clip ends. Motivated by this *temporal saliency prior*, we aggregate the per-frame predictions into a single clip-level confidence anchor p_{final} by retaining the last-frame prediction, which empirically exhibits the highest signal-to-noise ratio among all frame positions.

Temporal risk prior reconstruction. Accident risk possesses a well-known *monotonicity prior*: the probability of an imminent collision increases monotonically as the temporal distance to the event decreases. We exploit this inductive bias to reconstruct a dense, physically plausible frame-level risk curve from the clip-level anchor. Specifically, we parameterise the reconstruction as

$$p_t = p_{\text{start}} + (\beta\, p_{\text{final}} - p_{\text{start}}) \cdot h(t/T;\, k, \mu), \qquad (2)$$

where h is a monotonically increasing temporal basis function that blends exponential and logarithmic schedules, k and μ are shape parameters estimated from the second-order statistics of the raw curve, $\beta < 1$ is a head-room factor ensuring strict monotonicity below the end anchor, and p_{start} is the initial risk level. This step effectively *propagates* the model's high-confidence late-segment signal backward along the temporal axis under a physically motivated constraint.

Prediction distribution alignment. Even after temporal reconstruction, the predicted confidence distribution on the target domain remains systematically shifted relative to the source domain's decision boundary. Since no target-domain labels are available, we perform *unsupervised recalibration* using only the moments of the model's own prediction distribution on the target set. Concretely, we compute the empirical median m of $\{p_{\text{final}}\}$ across all target clips as a domain-specific statistic, and apply an order-preserving mapping that aligns the predicted distribution to the expected decision boundary. This procedure is analogous to test-time batch normalisation in unsupervised domain adaptation: it shifts the activation statistics to match the target distribution without retraining. Crucially, because the only target-set quantity used is a moment of the model's own output distribution, the zero-shot constraint is fully respected.

3. Experiments

3.1. Implementation Details

The full model (87.1 M parameters) is fine-tuned with AdamW [7], learning rate 1×10^{-5} for the backbone and 1×10^{-4} for the head, weight decay 5×10^{-4}, batch size 3 with 8-step gradient accumulation (effective batch 24), 20 epochs, StepLR ($\gamma = 0.1$ every 5 epochs), gradient clipping with max norm 5.0, and automatic mixed precision (AMP). Training takes approximately 17 hours on a single NVIDIA V100 GPU. We select the submitted checkpoint based on the validation AP, AUC, and mTTA.

Model scale. The VideoMAE-v2 Base backbone has 86.2 M parameters; the per-frame head adds 0.9 M, totalling 87.1 M trainable parameters.

Test-time adaptation hyper-parameters. On the source validation set, the raw model produces median $p_{\text{final}} \approx 0.34$ and clip-mean ≈ 0.05. The temporal risk prior uses $k \in [7, 15]$, $\mu \in [0.2, 0.8]$, and head-room factor $\beta = 0.998$. The distribution alignment step maps clips in $[m, 0.5)$ order-preservingly into $[0.505, 0.7]$.

3.2. Evaluation Protocol

We follow the official evaluation protocol of the CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition [1]. The evaluation measures detection quality via frame-level Average Precision (AP) and Area Under the ROC Curve (AUC), while early and stable risk awareness is assessed via Time-To-Accident (TTA) and Stable TTA (STTA). Higher AP and AUC indicate better detection capability, whereas larger TTA metrics reflect earlier anticipation. On our held-out Nexar [9] validation split, we report the AP and AUC metrics. Additionally, we present the competition's official weighted composite score evaluated on the unseen test set, denoted as "Private LB" in our results.

3.3. Results

Beyond the final pipeline of Sec. 2, we ran two experiments to test design choices that, *a priori*, looked promising but turned out to hurt the score. Both are reported here because their negative results directly motivate decisions in our final method. In both, the *only* change from the final pipeline is the one explicitly stated; everything else is held constant.

Late-Window Classification. Instead of feeding the model a 16-frame window over the full 5 s clip, we extract only the *last 2 s* of every clip, uniformly sub-sample 16 frames from this trailing window, and produce a single clip-level binary decision (rather than a per-frame risk sequence). The clip-level probability is then replicated across all 150 output positions. Trained with the same backbone and class-balanced split as the proposed model, this variant reaches a validation AP $= 0.66$ and AUC $= 0.69$ (Tab. 1, row B), clearly below the proposed per-frame model. On the private leaderboard the gap widens to -0.027 with respect to our final submission. The underlying reason is that, by replicating a single scalar across all 150 positions, the late-window output has no temporal resolution by construction, so the post-processor of Sec. 2.4—which is order-preserving—cannot rebuild a rising risk curve from a flat one. Empirically, this confirms that *discarding the first 3 s of context costs more than the simpler late-time supervision saves*; we therefore

	Method	AP	AUC	Private LB
A	Proposed	**0.83**	**0.86**	**2.46234**
B	Late-Window	0.66	0.69	2.43530
C	Mixed w/o balance	0.78	0.60	2.35424

Table 1. Comparisons on the Nexar validation split. "AP" and "AUC" are computed on the held-out Nexar validation set, with AP reported as the standard average precision in $[0, 1]$. "Private LB" reports the corresponding submission's score on the official Kaggle *private* leaderboard of the CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition, where higher is better.

retain the full 5 s input plus a per-frame head.

Cross-Dataset Mixed Training without Class Balancing. Motivated by the scarcity of accident clips in Nexar, we augment training with a second public accident dataset [22] and use the union as the new training set. Crucially, in this experiment we *bypass the class-balancing step of* Sec. 2.2: every positive clip from both sources is kept, and only Nexar's purely-negative videos are sub-sampled. The resulting training set contains substantially more positives than negatives. The model trained under this regime degenerates into an "always-positive" predictor: on validation it still attains AP $= 0.78$ (Tab. 1, row C), because AP on an imbalanced set rewards any model that tends to predict the majority class, but the corresponding AUC $= 0.60$—close to the 0.5 chance level—reveals that the model has essentially lost the ability to rank positives above negatives. The private-leaderboard score drops to 2.35, a full -0.108 gap below our final submission. This is a textbook consequence of class imbalance at the dataset level: when the marginal $P(y = 1)$ in training is pushed close to one, the cross-entropy minimiser is the constant-positive predictor, and the time-weighting term in the Exp-Loss—which only re-scales the *positive* loss—provides no counterweight. It is for exactly this reason that we make dataset-construction-time class balancing a first-class component of our pipeline rather than a tunable hyper-parameter.

4. Conclusion

We presented a zero-shot traffic accident anticipation framework that couples a VideoMAE-v2 sliding-window classifier with class-balanced dataset construction and a training-free test-time domain adaptation module. Ablation studies confirm that both full-context per-frame modelling and dataset-level class balancing are essential to avoid performance collapse. The system achieves 2nd place in the 2026 CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition [1].

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AutoMine Solution for AV2 2026 Scenario Mining Challenge

Songliang Cao[1,2]* Jiele Zhao[1]* Yuru Wang[1] Hao Li[1] Daqi Liu[1] Zehan Zhang[1]†
Fangzhen Li[1]† Yu Wang Yue Zhang Bing Wang[1] Guang Chen[1] Hao Lu[2] Hangjun Ye[1]
[1]Xiaomi EV [2]Huazhong University of Science and Technology

Abstract

With the development of autonomous driving systems, mining high-value, safety-critical, and planning-relevant scenarios from large-scale driving logs has become essential for data-driven evaluation. In this paper, we propose ***AutoMine****, a robust self-refining scenario mining method based on LLMs and VLMs. AutoMine uses semantics-preserving prompt augmentation to reduce LLM prompt sensitivity, combines robust trajectory atomic functions with VLM-based functions to handle perception noise and open-world visual cues, and refines generated code through execution feedback from real logs. In the Argoverse 2 Scenario Mining Competition at CVPR 2026, AutoMine achieves a HOTA-Temporal score of* ***36.38*** *and a Timestamp BA score of* ***77.21****.*

1. Introduction

Autonomous driving datasets contain massive sensor logs, while rare and safety-critical events remain sparse. Scenario mining enables targeted evaluation by retrieving logs, timestamps, and 3D actors that match a natural language description.

This task is challenging because query wording is precise, predicted tracks are noisy, and some scenarios require visual evidence beyond 3D trajectories. For example, *passing* and *overtaking* may imply different conditions, while tracks may contain missing detections, heading noise, fragmentation, and ID switches.

The rapid development of LLMs and VLMs brings new possibilities for scenario mining. RefProg [1] shows that LLMs can translate natural language descriptions into composable atomic function calls for interpretable trajectory-based mining. However, manually specified atomic functions are hard to scale to open-world visual concepts. In addition, one-shot generated code can fail when the LLM misunderstands the prompt, selects the wrong category, reverses a relation, or imposes incorrect constraints; this is consistent with prior findings on LLM sensitivity to meaning-preserving prompt design choices [3].

We propose **AutoMine**, a multimodal scenario mining framework that strengthens LLM-generated programs with semantic-preserving prompt augmentation, robust atomic functions, VLM-based visual functions, perception post-processing, and execution-driven self-refinement. AutoMine executes generated code on real logs, summarizes the observed outputs, and uses the feedback to repair systematic errors, improving robustness to language ambiguity and perception noise.

2. Method

2.1. Overview

Given a natural language description, driving logs, and sensor data, AutoMine outputs referred actors and valid timestamps. As shown in Fig.1, AutoMine first refines perception tracks, then combines semantic-preserving prompt augmentation, LLM-generated atomic functions, multimodal execution over tracks, maps, and images, and execution-feedback-based code refinement. We describe each module in the following sections.

2.2. Trajectory Refinement

AutoMine uses the detection and tracking results from Le3DE2E [5] as initial trajectory inputs. Although these results provide strong 3D tracks, we observe ID switches, fragmented trajectories, missed detections, false positives, and duplicated boxes, which directly affect temporal localization and referred-actor selection.

Inspired by Immortal Tracker [4], we keep short unmatched tracklets alive instead of terminating them immediately, and reconnect compatible fragments using spatial, temporal, category, and size consistency. We also apply backward tracking to recover early segments that are easier to associate from later frames. In addition, we use the grounding capability of Qwen3.5-27B [2] to verify projected boxes in camera views and remove additional false positives. This trajectory refinement improves track con-

*This work was done during internship at Xiaomi.
†Project leader.

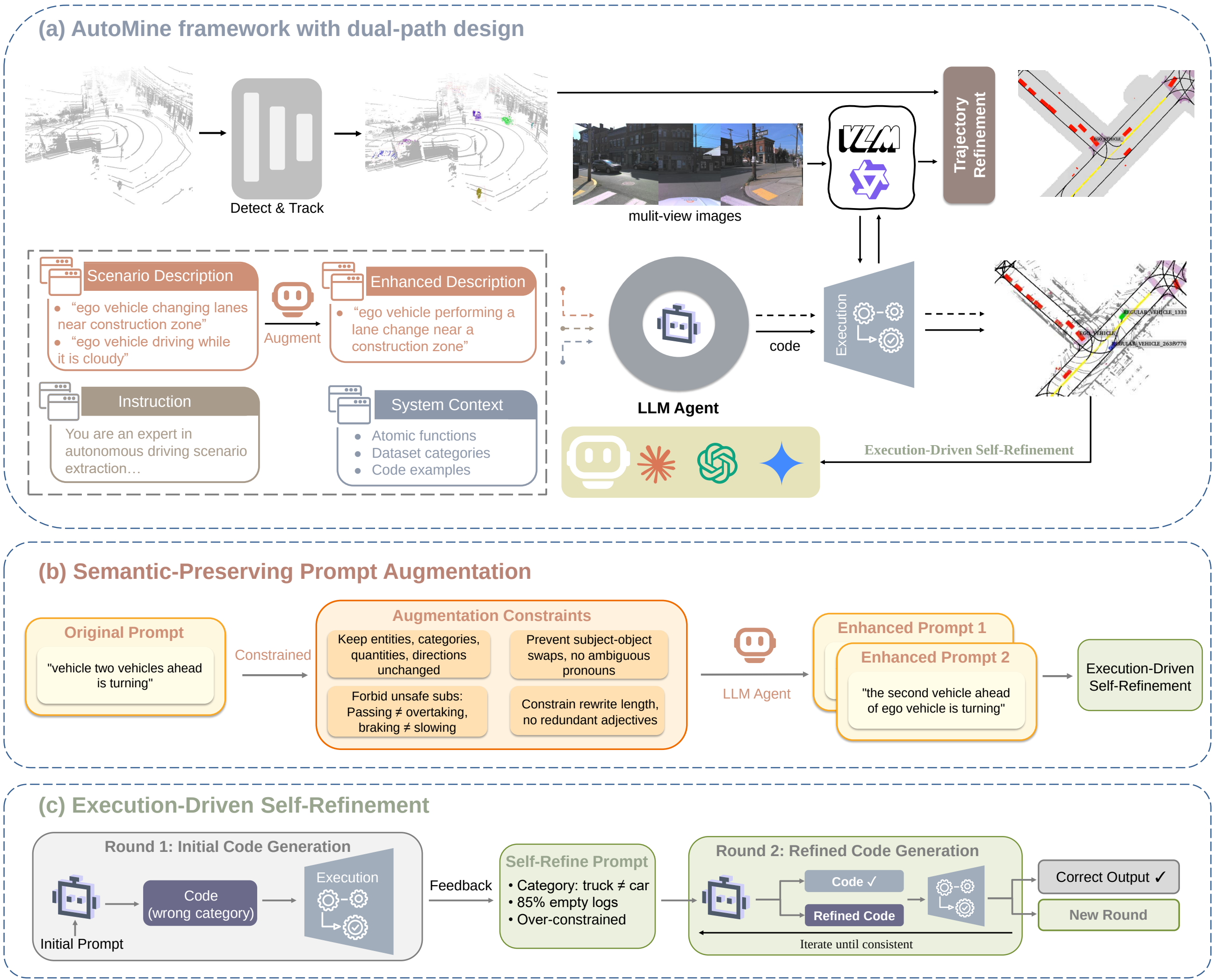

Figure 1. (a): Overview of the AutoMine framework with dual-path design (perception + language). (b): Semantic-preserving prompt augmentation. (c): Execution-driven self-refinement loop.

tinuity and provides more reliable inputs for downstream atomic functions.

2.3. Semantic-Preserving Prompt Augmentation

As mentioned above, prior work shows that LLMs are sensitive to prompt wording and formatting [3], and we observe the same issue when generating scenario mining code. To reduce this instability, AutoMine augments the natural language scenario descriptions before code generation.

The augmentation is constrained to preserve the original semantics rather than freely paraphrase the query. The rewrite prompt explicitly keeps all entities, categories, quantities, directions, road context, spatial-temporal relations, numerical values, and the referred target unchanged. It also forbids unsafe substitutions such as *passing* $\neq$ *overtaking*, *braking* $\neq$ *slowing*, *changing lanes* $\neq$ *merging*, *stopped* $\neq$ *parked*, and *near* $\neq$ *next to*. To avoid misleading the LLM's judgment of referred objects, related objects, and relation directions, we further enforce constraints that prevent subject-object swaps, demoting the referred actor into a modifier, changing category granularity, or introducing ambiguous pronouns. We also constrain the rewrite length to prevent over-introducing redundant adjectives or extra background.

2.4. Robust Trajectory-Based Atomic Functions

AutoMine represents scenario logic as compositions of atomic functions over tracks, ego poses, maps, and time windows. Since predicted tracks are noisy, we refactor motion and relation functions to use temporally aggregated evidence instead of brittle single-frame measurements. Directional and spatial relations are evaluated over multiple valid frames, with relaxed continuity checks and ego-relative geometry.

We also extend the library for planning-centric behaviors not covered by the baseline functions, including U-turns, three-point turns, side parking, special stopping behavior, object interactions, and map-aware road constraints. We use LLM clustering over all descriptions to identify missing function categories, then manually revise and implement the final function set.

2.5. VLM-Enhanced Atomic Functions

Some scenarios require visual evidence beyond 3D tracks and maps. AutoMine therefore provides VLM-enhanced atomic functions for fine-grained object type, visual attributes, environment, road surface, zones, pedestrian actions, traffic lights, occlusion, and attached objects. We use Qwen3.5-27B [2] as the underlying vision-language model for all visual reasoning calls.

For visual reasoning about candidate actors, we collect all camera views where the candidate appears and draw the projected 3D box on the original image instead of cropping it, preserving context under projection noise. For environment-level conditions, we use a representative front-view image because each log is short and the environment is usually stable. For road and zone conditions, we stitch candidate-visible frames into a panel annotated with camera names and timestamp indices, and the VLM returns the valid timestamps. Ego-related visual functions use separate prompts because the ego vehicle is not represented by a normal projected object box.

The VLM-enhanced function library includes the functions in Table 1. These functions allow AutoMine to preserve the advantages of symbolic program execution while extending coverage to open-world visual concepts.

Table 1. VLM-enhanced atomic functions.

Function	Description
`is_specific_object_type()`	Identifies fine-grained object types.
`in_environment()`	Checks weather or environment conditions.
`has_attribute()`	Verifies visual attributes or states.
`on_road_type()`	Determines the road type.
`on_road_surface()`	Checks road surface conditions.
`in_zone()`	Determines whether an actor is in a visual zone.
`pedestrian_action()`	Recognizes pedestrian actions.
`object_color()`	Identifies object color.
`at_traffic_lights()`	Checks traffic-light-related conditions.
`is_occluded_by()`	Detects visual occlusion relations.
`has_attached_object()`	Checks attached or carried objects.

2.6. Execution-Driven Self-Refinement

LLM-generated mining code often fails in systematic ways, such as selecting the wrong referred category, reversing relation-function arguments, missing `reverse_relationship`, using overly strict thresholds, or misjudging front/back and left/right geometry. Since these errors are hard to identify from code alone, AutoMine refines code with feedback from real execution.

In each round, a code generator produces scenario-mining code, and an executor runs it on up to `max_logs` logs with trajectory and VLM atomic functions. AutoMine then summarizes what the code retrieves, including per-track category, temporal coverage, size, ego-object geometry, and cross-log statistics such as referred-category distribution, empty-log ratio, and related-object distribution. These diagnostics expose category confusion, noisy short tracks, over-constrained logic, and relation-direction errors.

The next refinement prompt includes the original description, function library, category definitions, previous code, structured feedback, and the referred category from `REFERRED_DICT` as a hard constraint. The LLM keeps the code unchanged if the feedback is consistent; otherwise, it repairs function choices, argument order, reverse relations, thresholds, category filters, or spatial reasoning. This forms an execution-grounded self-refinement loop.

3. Experiments

3.1. Dataset and Evaluation Metrics

We conduct experiments on the official benchmark of the CVPR 2026 Argoverse 2 Scenario Mining Challenge. The benchmark is built upon the Argoverse 2 Sensor Dataset [6], which contains 1,000 driving logs (700 training, 150 validation, 150 test) with a total of approximately 4.2 hours of driving data. Each log lasts about 15 seconds. The sensor suite includes two 32-beam LiDARs (10 Hz), nine global shutter cameras (20 fps), HD maps, and 6-DOF ego-vehicle poses. The dataset provides 10,000 planning-centric natural language queries. The evaluation metrics are as follows:

HOTA-Temporal ($\uparrow$). The primary ranking metric, computed only on the time window where the scenario description holds. For each prompt, predictions are filtered by both class and a per-prompt confidence threshold (selected from 10 recall-based candidates), and the standard HOTA score is computed using center-distance similarity (zero-distance $= 2\,\text{m}$). HOTA jointly measures detection and association

Table 2. Ablation results on the Argoverse 2 validation set.

Configuration	HOTA-Temporal(↑)	HOTA-Track(↑)	Timestamp BA(↑)	Log BA(↑)
Different LLM backends (single-query baseline)				
gpt-5.3-codex-9	15.72	24.93	61.19	63.76
Gemini-2.5-Pro	21.32	29.54	64.49	64.95
Claude-Sonnet-4.6	22.14	31.05	67.17	67.97
Ablations on Claude-Sonnet-4.6				
+ Trajectory Refinement	24.04	35.96	67.51	71.59
+ Atomic Function Optimization (Robust + VLM)	31.46	41.86	74.61	76.98
+ Execution-Driven Self-Refinement	33.96	44.95	75.98	**80.04**
+ Semantic-Preserving Prompt Augmentation	**34.99**	**45.91**	**77.87**	79.83

Table 3. Official leaderboard results of the AV2 2026 Scenario Mining Challenge on the HOTA-Temporal track.

Rank	Team	HOTA-Temporal(↑)	HOTA-Track(↑)	Timestamp BA(↑)	Log BA(↑)
1	HYU_OASIS	**38.50**	52.63	74.32	77.12
2	MTL (Argonaut)	37.04	**55.11**	75.50	**80.75**
3	**MISI (AutoMine) (Ours)**	36.38	49.32	**77.21**	76.26
4	Zelos_Agent_Driving	33.45	45.23	72.45	73.66
5	Lilly	32.51	42.06	71.93	73.11
6	DataClaw	31.40	44.40	74.05	79.10

quality:

$$\begin{cases} \mathrm{HOTA}_\alpha = \sqrt{\mathrm{DetA}_\alpha \cdot \mathrm{AssA}_\alpha}, \\ \mathrm{HOTA} = \dfrac{1}{|A|} \displaystyle\sum_{\alpha \in A} \mathrm{HOTA}_\alpha, \end{cases} \tag{1}$$

where $A = \{0.05, 0.10, \ldots, 0.95\}$ is the set of localization thresholds, and DetA, AssA denote detection and association accuracy, respectively. The final score per prompt is the maximum HOTA over the 10 candidate thresholds.

HOTA-Track (↑). Same as HOTA-Temporal, except that any track ever marked as `REFERRED_OBJECT` in a sequence is treated as referred across all of its frames. This evaluates the full lifetime of referred tracks rather than only the description-active interval.

Timestamp BA (↑). A frame-level retrieval metric that asks whether any referred object exists in each timestamp, independent of localization. Using the optimal thresholds from HOTA-Temporal, each frame is classified as positive iff it contains at least one `REFERRED_OBJECT`. Per prompt, we aggregate frame-level TP/FP/TN/FN and compute balanced accuracy:

$$\text{Timestamp BA} = \frac{1}{2}\left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} + \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}\right). \tag{2}$$

Log BA (↑). The log-level counterpart of Timestamp BA. A (log, prompt) sequence is positive iff *any* frame contains a referred object. BA is then computed across all sequences using the same formulation as above.

3.2. Ablation Study

To validate the contribution of each design in AutoMine, we conduct ablation studies on the validation set, using the publicly available Le3DE2E [5] trajectories as initial input. We first compare three state-of-the-art LLMs under the single-query baseline and adopt Claude-Sonnet-4.6 as the default code generator, since it is the most stable on relation-argument ordering and fine-grained category selection. For all VLM-enhanced atomic functions, we use Qwen3.5-27B [2] as the underlying vision-language model. We then incrementally add each component on top of this baseline. The results are shown in Table 2.

We observe that each component improves the system through a distinctly different mechanism rather than simply boosting all metrics uniformly. Trajectory refinement contributes a much larger HOTA-Track gain than HOTA-Temporal gain, indicating that re-linking fragmented tracklets mainly extends the lifetime of already-correct referred objects rather than enlarging the temporal window of new scenarios. Atomic function optimization brings the largest single jump, because temporally aggregated relation predicates absorb per-frame heading noise on directional queries, while VLM-enhanced functions cover attributes that are unobservable from 3D boxes alone (e.g., traffic-light state, road surface, attached cargo); the two are largely complementary and fail on disjoint subsets of queries.

Execution-driven self-refinement further improves

Table 4. Official leaderboard results of the AV2 2026 Scenario Mining Challenge on the Timestamp BA track.

Rank	Team	Timestamp BA(↑)	Log BA(↑)
1	**AutoMine (Ours)**	**77.21**	76.26
2	MTL (Argonaut)	75.50	**80.75**
3	HYU_OASIS	74.69	77.84
4	DataClaw	74.18	78.67
5	EABOT_AMD	72.86	71.09
6	Lilly	72.70	75.77

all metrics by repairing three recurring error patterns we observe in the round-0 code: missing `reverse_relationship` calls, over-strict thresholds that yield zero candidates, and wrong referred categories caught against `REFERRED_DICT`. Semantic-preserving prompt augmentation yields only marginal gains on the raw baseline but becomes effective once stacked on top of the optimized pipeline, since it reduces prompt-induced variance across semantically equivalent rewrites rather than introducing new capability—an effect that is naturally amplified when the downstream pipeline is already strong.

3.3. Leaderboard Results

Our final submission, AutoMine, achieves excellent results on the test set of the CVPR 2026 Argoverse 2 Scenario Mining Challenge. Table 3 shows the leaderboard sorted by the primary metric HOTA-Temporal: we rank **3rd** with **36.38** and HOTA-Track of 49.32. Table 4 shows the leaderboard sorted by Timestamp BA: we rank **1st** with **77.21**, demonstrating the advantage of AutoMine in temporal localization accuracy.

4. Conclusion

In this report, we presented **AutoMine**, a robust multimodal scenario mining framework for the AV2 2026 Scenario Mining Challenge. AutoMine addresses several key challenges in natural-language-driven scenario mining, including LLM prompt sensitivity, noisy and fragmented perception tracks, ambiguous spatial-temporal relations, and open-world visual concepts that cannot be captured by 3D trajectories alone. To this end, AutoMine refines raw trajectories, applies semantics-preserving prompt augmentation, builds robust trajectory-based and VLM-enhanced atomic functions, and further improves generated mining programs through execution-driven self-refinement using feedback from real logs. Experiments and ablation studies on the Argoverse 2 benchmark show that these components provide complementary gains, with atomic function optimization and self-refinement substantially improving actor retrieval and temporal localization. On the official leaderboard, AutoMine achieves the best Timestamp BA score of 77.21, demonstrating strong temporal localization ability, and ranks 3rd in HOTA-Temporal with a score of 36.38. These results show the promise of integrating symbolic programs, perception refinement, and visual reasoning for scenario mining.

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